

JADAVPUR UNIVERSITY

MCA - 1ST YEAR 2ND SEMESTER 2024

LAB ASSIGNMENT - DATABASE MANAGEMENT SYSTEM

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Online Dairy Product Management System

General Description:

The Online Product Management System for dairy products is a sophisticated platform designed to revolutionize the way perishable dairy items are managed and delivered to customers. In addition to core features such as inventory management and delivery tracking, the system integrates a secure and reliable payment gateway to facilitate seamless transactions.

The payment gateway feature allows customers to securely complete their purchases online, enhancing convenience and trust in the system.

Problem Statement:

- **Inefficient Product Management:** Without the system, manual tracking of dairy products becomes prone to errors, leading to inaccuracies in stock levels, expiration dates, and shelf life management.
- **Delivery Delays and Inconsistencies:** Lack of optimization algorithms may result in inefficient delivery routes, leading to delays in product delivery and inconsistencies in service quality.
- **Increased Product Wastage:** Absence of real-time shelf life monitoring may lead to higher instances of product wastage as expired items may go unnoticed, impacting profitability and sustainability.
- **Poor Customer Experience:** Customers may experience inconvenience due to the absence of live tracking features, leading to uncertainty about the status of their orders and dissatisfaction with the service.
- **Manual Transaction Processing:** Manual handling of transactions increases the risk of errors and security breaches, potentially exposing sensitive customer information to unauthorized access.
- **Limited Scalability:** Manual processes limit the scalability of the business, making it challenging to expand operations and meet growing customer demands efficiently.

System Requirement & Analysis:

Non-Functional Requirements:

- **Performance:**

Response Time: The system should respond to user actions within a maximum of 3 seconds under normal load conditions.

Throughput: The system should support a minimum of 1000 simultaneous users without experiencing degradation in performance.

Scalability: The system should be capable of handling a 20% increase in user load without significant impact on performance.

- **Reliability:**

Availability: The system should have at least 99.9% uptime, excluding scheduled maintenance periods.

Fault Tolerance: The system should continue to operate seamlessly in the event of hardware or software failures, with minimal disruption to users.

Data Integrity: Data stored in the system should be accurate and consistent, with mechanisms in place to prevent data corruption or loss.

- **Security:**

Data Encryption: All sensitive data, including user credentials, payment information, and personal details, should be encrypted both in transit and at rest.

Authentication and Authorization: The system should employ robust authentication mechanisms (e.g., multi-factor authentication) to ensure that only authorized users can access sensitive functionalities.

Access Control: Role-based access control (RBAC) should be implemented to restrict access to certain features or data based on user roles and permissions.

Secure Payment Processing: The payment gateway should comply with industry standards (e.g., PCI DSS) to ensure secure handling of payment transactions and protection against fraud.

- **Usability:**

Intuitive User Interface: The user interface should be user-friendly and intuitive, allowing users to navigate the system easily and perform tasks without extensive training.

Accessibility: The system should be accessible to users with disabilities, following relevant accessibility standards (e.g., WCAG).

Multi-platform Compatibility: The system should be compatible with major web browsers and mobile devices to accommodate users accessing the platform from different devices.

- **Scalability:**

Horizontal Scalability: The system architecture should support horizontal scaling to accommodate increased user load or data volume by adding more resources or servers.

Vertical Scalability: The system should support vertical scaling to handle increased processing demands by upgrading hardware components (e.g., CPU, memory) without significant downtime.

- **Maintainability:**

Modularity: The system should be designed with modular components to facilitate easier maintenance and updates without impacting the entire system.

Documentation: Comprehensive documentation should be provided for system architecture, codebase, APIs, and deployment procedures to aid in maintenance and troubleshooting.

Logging and Monitoring: The system should incorporate logging and monitoring mechanisms to track system behaviour, detect anomalies, and facilitate debugging and performance optimization.

- **Compliance:**

Regulatory Compliance: The system should comply with relevant regulations and standards governing data privacy, security, and online transactions (e.g., GDPR, HIPAA, PSD2).

Industry Standards: The system should adhere to industry best practices and standards in software development, security, and quality assurance.

Functional Requirements:

- **User Authentication and Authorization:**

Users should be able to register and log in to the system using unique credentials (username/email and password).

The system should authenticate user credentials securely and grant appropriate access levels based on user roles (e.g., customer, delivery personnel, administrator).

- **Real-time Product Process Tracking:**

Users should be able to track the status of their orders in real-time, from placement to delivery.

The system should provide live updates on the processing stages of orders, including order confirmation, preparation, packaging, and delivery.

- **Product Management:**

Administrators should be able to add new products to the system by providing details such as product name, description, price, and expiration date (if applicable).

- **Inventory Management:**

The system should maintain accurate records of available stock for each product, including current quantity and location within the inventory.

Inventory levels should be automatically updated in real-time.

- **Online Payment and Invoice Generation:**

Customers should be able to select products for purchase and proceed to checkout.

The system should securely process online payments using various payment methods, including credit/debit cards, digital wallets, and other supported options.

- **Order Management:**

Customers should have the ability to view their order history, including past purchases and order statuses.

Administrators should be able to manage orders, including order processing, fulfilment, and cancellation if necessary.

SOFTWARE AND HARDWARE ANALYSIS AND REQUIREMENTS:

- SOFTWARE:

Any operating system for we browse interface

Database Oracle Database is used as database as it easy to maintain and retrieve records by simple queries which are in English language which are easy to understand and easy to write

- HARDWARE:

At least 1GB storage on the server side

Internet connection on both reader and server side

Data Collection:

- User Table

id	name	email	password	role	createdAt
1	Soumen Gharui	soumen@example.co m	****	admin	2024-01-15 10:00:00
2	Sohini Das	Sohini@example.com	****	user	2024-01-16 11:00:00
3	Arka Das	Arka@example.com	****	user	2024-02-17 09:30:00
4	Subrata Das	subrata@example.com	****	user	2024-03-18 08:45:00

- Product Table

id	name	description	price	stock	category	ratings	images	expiry_date	created_at	admin_id
1	Milk	Fresh cow milk	1.50	100	Dairy	4.5	milk.png	2023-12-01	2023-01-15 10:00:00	1
2	Cheese	Cheddar cheese	3.00	50	Dairy	4.7	cheese.png	2024-01-10	2023-01-15 10:00:00	1
3	Yogurt	Greek yogurt	2.00	200	Dairy	4.8	yogurt.png	2023-11-20	2023-02-10 11:00:00	1
4	Butter	Salted butter	4.00	30	Dairy	4.6	butter.png	2023-10-15	2023-02-10 11:00:00	1

- Cart Table

id	product_id	name	price	quantity
1	1	Milk	1.50	2
2	2	Cheese	3.00	1
3	3	Yogurt	2.00	3
4	4	Butter	4.00	1

- Orders Table

id	user_id	total_price	order_status	created_at	paymentinfo	shipping_info_id
1	1	6.00	Completed	2023-01-20 12:00:00	Paid	1
2	2	3.00	Pending	2023-01-21 13:00:00	Unpaid	2
3	3	12.00	Completed	2023-02-22 14:00:00	Paid	3
4	4	4.00	Shipped	2023-03-23 15:00:00	Paid	4

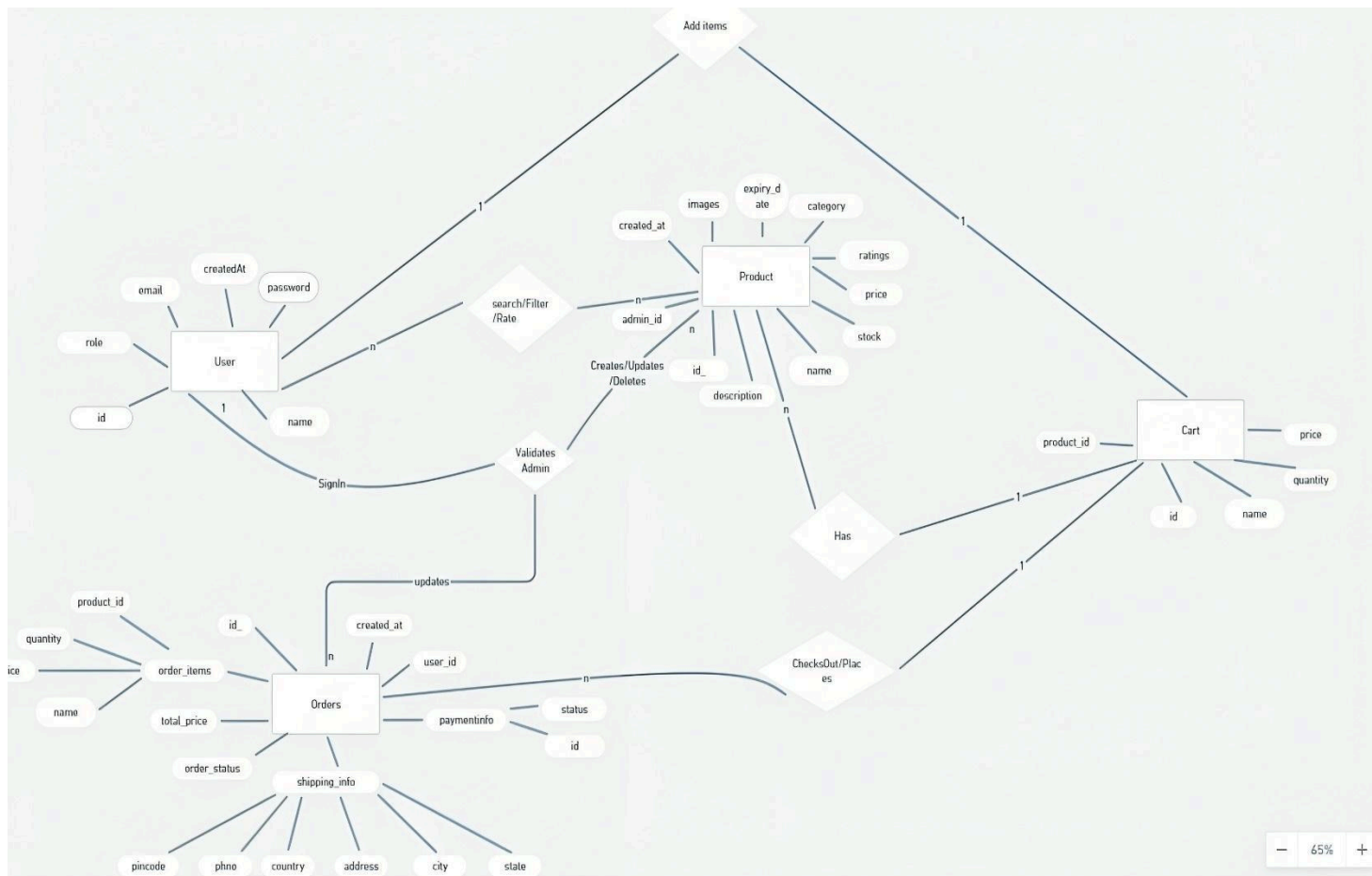
- Order Items Table

id	order_id	product_id	quantity	price
1	1	1	2	3.00
2	2	2	1	3.00
3	3	3	3	6.00
4	4	4	1	4.00

- **Shipping Info Table**

id	address	city	state	country	pincode	phno
1	Baghajatin	Kolkata	WB	India	12345	986446461
2	2/3 Chittaranjan Street	Kolkata	WB	India	67890	9754613566
3	B.C Road	Burdwam	WB	India	713101	963258711
4	Riksha Stand	Rishra	WB	India	762214	974125893

ER Diagram



ASSIGNMENT 1

1.Create the following tables :

Tablename: EMPLOYEE

Structure :

EMP_CODE	char(16)
EMP_NAME	char(20)
DEPT_CODE	char(16)
DESIG_CODE	char(16)
SEX	char(1)
ADDRESS	char (25)
CITY	char (20)
STATE	char (20)
PIN	char (6)
BASIC	Number
JN_DT	Date

Primary key is EMP_CODE

Tablename : DESIGNATION

Structure :

DESIG_CODE	char(16)
DESIG_DESC	char(20)

Primary key is DESIG_CODE

Tablename : DEPARTMENT

Structure :

DEPT_CODE	char(16)
DEPT_NAME	char(20)

Primary key is DEPT_CODE.

create table EMPLOYEE(
EMP_CODE char(5) primary key,

```

EMP_NAME char(20),
DEPT_CODE char(5),
DESIG_CODE char(5),
SEX char(1),
ADDRESS char(25),
CITY char(20),
STATE char(20),
PIN char(6),
BASIC Number,
JN_DT date);

```

```

CREATE TABLE DESIGNATION(
DESIG_CODE CHAR(5) PRIMARY KEY,
DESIG_DESE CHAR(20)
)

```

```

CREATE TABLE DEPARTMENT(
DEPT_CDOE CHAR(5) PRIMARY KEY,
DEPT_NAME CHAR(15)
)

```

2. Display the structure of each table .

TABLE EMPLOYEE

Column	Null?	Type
EMP_CODE	NOT NULL	CHAR (5)
EMP_NAME	-	CHAR (20)
DEPT_CODE	-	CHAR (5)
DESIG_CODE	-	CHAR (5)
SEX	-	CHAR (1)
ADDRESS	-	CHAR (25)
CITY	-	CHAR (20)
STATE	-	CHAR (20)
PIN	-	CHAR (6)

BASIC	-	NUMBER
JN_DT	-	DATE

TABLE DESIGNATION

Column	Null?	Type
DESIG_CODE	NOT NULL	CHAR(5)
DESIG_DESE	-	CHAR(20)

TABLE DEPARTMENT

Column	Null?	Type
DEPT_CODE	NOT NULL	CHAR(5)
DEPT_NAME	-	CHAR(15)

3. Insert a few rows in each table .[While entering data in the EMP table use DESIG_CODE which exists in the DESIGNATION table and DEPT_CODE which exists in the DEPARTMENT table. In the DESIGNATION table, assign code for Manager, Executive, officer, clerk and helper. In DEPARTMENT table, assign code for Personnel, Production, Purchase, Finance, Research departments]

```
insert into DESIGNATION values ('D001','Manager');
```

```
insert into DESIGNATION values ('D002','Executive');
```

```
insert into DESIGNATION values ('D003','Officer');
```

```
insert into DESIGNATION values ('D004','Clerk');
```

```
insert into DESIGNATION values ('D005','Helper');
```

insert into DEPARTMENT values ('DP01','Personnel');

insert into DEPARTMENT values ('DP02','Production');

insert into DEPARTMENT values ('DP03','Purchase');

insert into DEPARTMENT values ('DP04','Finnance');

insert into DEPARTMENT values ('DP05','Research ');

insert into EMPLOYEE values ('E001','TAPAN','DP01', 'D001', 'M',
'BALARAMPUR','PURULIA','WESTBENGAL','723103',80000,
to_date('21:01:2024','dd:mm:yyyy'));

insert into EMPLOYEE values ('E002','ROHIT','DP02', 'D002', 'M', 'ABC','PURULIA','WEST
BENGAL','723102',70000, to_date('21:01:2024','dd:mm:yyyy'));

insert into EMPLOYEE values ('E004','AMIT','DP03', 'D003', 'M', 'ABC','PURULIA','WEST
BENGAL','723102',0,
to_date('21:01:2024','dd:mm:yyyy'));

insert into EMPLOYEE values ('E005','ABC','DP03', 'D004', 'M', 'ABC_D','PURULIA','WEST
BENGAL','723102',0, to_date('01:01:2024','dd:mm:yyyy'));

insert into EMPLOYEE values ('E006','DEF','DP02', 'D005', 'M', 'ABC','PURULIA','WEST
BENGAL','723102',0, to_date('02:01:2024','dd:mm:yyyy'));

insert into EMPLOYEE values ('E007','GHI','DP01', 'D005', 'M', 'ABC','PURULIA','WEST
BENGAL','723102',0, to_date('22:01:2024','dd:mm:yyyy'));

insert into EMPLOYEE values ('E008','ANITA','DP01', 'D005', 'F', 'ABC','PURULIA','WEST
BENGAL','723102',60000, to_date('23:01:2024','dd:mm:yyyy'));

insert into EMPLOYEE values ('E009','RITA','DP02', 'D002', 'F', 'ABC','KOLKATA','WEST
BENGAL','723102',70000, to_date('21:01:2023','dd:mm:yyyy'));

insert into EMPLOYEE values ('E010','ANKITA','DP04', 'D003', 'F', 'ABC','KOLKATA','WEST
BENGAL','700000',95000, to_date('22:05:2023','dd:mm:yyyy'));

insert into EMPLOYEE values ('E011','MARY','DP03', 'D005', 'F', 'ABC','KOLKATA','WEST
BENGAL','700000',65000, to_date('22:08:2022','dd:mm:yyyy'));

insert into EMPLOYEE values ('E012','MARY','DP01', 'D005', 'F', 'ABC','KOLKATA','WEST
BENGAL','700000',69000, to_date('23:08:2023','dd:mm:yyyy'));

insert into EMPLOYEE values ('E013','EDNA','DP01', 'D005', 'F', 'ABC','KOLKATA','WEST
BENGAL','700000',69000, to_date('23:08:2000','dd:mm:yyyy'));

```
insert into EMPLOYEE values ('E014','EDIE','DP01', 'D005', 'F', 'ABC','KOLKATA','WEST BENGAL','900000',69000, to_date('2:01:2000','dd:mm:yyyy'));
```

```
insert into EMPLOYEE values ('E015','MAX','DP02', 'D003', 'M', 'ABC','KOLKATA','WEST BENGAL','900000',69000, to_date('2:09:2000','dd:mm:yyyy'));
```

```
insert into EMPLOYEE values ('E016','JAMES','DP02', 'D002', 'M', 'ABC','KOLKATA','WEST BENGAL','900000',60000, to_date('2:09:2000','dd:mm:yyyy'));
```

```
insert into EMPLOYEE values ('E017','JAMIE','DP04', 'D002', 'M', 'ABC','KOLKATA','WEST BENGAL','900000',60000, to_date('24:09:2000','dd:mm:yyyy'));
```

```
insert into EMPLOYEE values ('E018','DONA','DP03', 'D001', 'F', 'ABC','KOLKATA','WEST BENGAL','700000',89000, to_date('21:09:2000','dd:mm:yyyy'));
```

4. In the EMP table insert row without DEPT_CODE and BASIC.

```
insert into EMPLOYEE values ('E003','SUDIP',NULL, 'D002', 'M', 'ABC','PURULIA','WEST BENGAL','723102', NULL, to_date('21:01:2024','dd:mm:yyyy'));
```

DESIG_CODE	DESIG_DESE
D001	Manager
D002	Executive
D003	Officer
D004	Clerk
D005	Helper

DEPT_CDOE	DEPT_NAME
DP01	Personnel

DP02	Production
DP03	Purchase
DP04	Finnance
DP05	Research

EMP_C ODE	EMP_N AME	DEPT_ CODE	DESIG_ CODE	SE X	ADDRES S	CITY	STATE	PIN	BAS IC	JN_ DT
E001	TAPAN	DP01	D001	M	BALARA MPUR	PURU LIA	WEST BENGAL	723 103	800 00	21- JAN -24
E002	ROHIT	DP02	D002	M	ABC	PURU LIA	WEST BENGAL	723 102	700 00	21- JAN -24
E003	SUDIP	-	D002	M	ABC	PURU LIA	WEST BENGAL	723 102	-	21- JAN -24
E004	AMIT	DP03	D003	M	ABC	PURU LIA	WEST BENGAL	723 102	0	21- JAN -24
E005	ABC	DP03	D004	M	ABC_D	PURU LIA	WEST BENGAL	723 102	0	01- JAN -24

E006	DEF	DP02	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	02- JAN -24
E007	GHI	DP01	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	22- JAN -24
E008	ANITA	DP01	D005	F	ABC	PURU LIA	WEST BENGAL	723 102	600 00	23- JAN -24
E009	RITA	DP02	D002	F	ABC	KOLK ATA	WEST BENGAL	723 102	700 00	21- JAN -23
E010	ANKIT A	DP04	D003	F	ABC	KOLK ATA	WEST BENGAL	700 000	950 00	22- MAY -23
E011	MARY	DP03	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	650 00	22- AUG -22
E012	MARY	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	690 00	23- AUG -23
E013	EDNA	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	690 00	23- AUG -00
E014	EDIE	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	900 000	690 00	02- JAN -00

E015	MAX	DP02	D003	M	ABC	KOLK ATA	WEST BENGAL	900 000	690 00	02- SEP -00
E016	JAMES	DP02	D002	M	ABC	KOLK ATA	WEST BENGAL	900 000	600 00	02- SEP -00
E017	JAMIE	DP04	D002	M	ABC	KOLK ATA	WEST BENGAL	900 000	600 00	24- SEP -00
E018	DONA	DP03	D001	F	ABC	KOLK ATA	WEST BENGAL	700 000	890 00	21- SEP -00

5. Find the rows with unassigned DEPT_CODE

SELECT * FROM EMPLOYEE WHERE DEPT_CODE IS NULL;

EMP_C ODE	EMP_N AME	DEPT_C ODE	DESIG_ CODE	SE X	ADDRE SS	CITY	STAT E	PIN	BAS IC	JN_ DT
E003	SUDIP	-	D002	M	ABC	PURU LIA	WEST BENG AL	7231 02	-	21- JAN -24

6. Find the rows with BASIC equal to zero

SELECT * FROM EMPLOYEE WHERE BASIC =0;

EMP_C ODE	EMP_N AME	DEPT_C ODE	DESIG_ CODE	SE X	ADDRE SS	CITY	STAT E	PIN	BAS IC	JN_ DT
E004	AMIT	DP03	D003	M	ABC	PURU LIA	WEST BENG AL	7231 02	0	21- JAN -24
E005	ABC	DP03	D004	M	ABC_ D	PURU LIA	WEST BENG AL	7231 02	0	01- JAN -24
E006	DEF	DP02	D005	M	ABC	PURU LIA	WEST BENG AL	7231 02	0	02- JAN -24
E007	GHI	DP01	D005	M	ABC	PURU LIA	WEST BENG AL	7231 02	0	22- JAN -24

7. 7. Find the rows with unassigned Basic [note down the difference between the results obtained in Q.6 and Q.7]

SELECT * FROM EMPLOYEE WHERE BASIC IS NULL;

EMP_C ODE	EMP_N AME	DEPT_C ODE	DESIG_ CODE	SE X	ADDRE SS	CITY	STAT E	PIN	BAS IC	JN_ DT
E003	SUDIP	-	D002	M	ABC	PURU LIA	WEST BENG AL	7231 02	-	21- JAN -24

8. Find the average basic of the employees

AVERAGE_BASIC
54411.7647058823529411764705882352941176

9. Replace the BASIC with 0 for the rows with unassigned Basic.

```
UPDATE EMPLOYEE SET BASIC = 0 WHERE BASIC IS NULL;
```

10. Again, find the average Basic. (Note the difference of result obtained in Q.8 & Q.10.)

AVERAGE_BASIC
51388.888888888888888888888888888888889

11. Delete the rows with unassigned DEPT_CODE.

```
DELETE FROM EMPLOYEE WHERE DEPT_CODE IS NULL;
```

12. Say, Net pay of an employee= Basic+HRA+DA where HRA is 50% of the Basic & DA is 40% of Basic. Show the employee name & Net pay for all employees.

EMP_NAME	BASIC	HRA	DA	NET_PAY
----------	-------	-----	----	---------

TAPAN	80000	40000	32000	152000
ROHIT	70000	35000	28000	133000
AMIT	0	0	0	0
ABC	0	0	0	0
DEF	0	0	0	0
GHI	0	0	0	0
ANITA	60000	30000	24000	114000
RITA	70000	35000	28000	133000
ANKITA	95000	47500	38000	180500
MARY	65000	32500	26000	123500
MARY	69000	34500	27600	131100
EDNA	69000	34500	27600	131100
EDIE	69000	34500	27600	131100
MAX	69000	34500	27600	131100

JAMES	60000	30000	24000	114000
JAMIE	60000	30000	24000	114000
DONA	89000	44500	35600	169100

13. Show the EMP_NAME & BASIC in the ascending order of DEPT_CODE. The employee name must appear in uppercase.

```
SELECT UPPER(EMP_NAME) AS EMPLOYEE_NAME , BASIC
FROM EMPLOYEE
ORDER BY DEPT_CODE ASC;
```

EMPLOYEE_NAME	BASIC
MARY	69000
EDIE	69000
ANITA	60000
EDNA	69000
TAPAN	80000
GHI	0
RITA	70000
DEF	0
JAMES	60000
ROHIT	70000

MAX	69000
ABC	0
AMIT	0
DONA	89000
MARY	65000
JAMIE	60000
ANKITA	95000

14. Find the employees who have joined after 1st January 2010.

```
SELECT *
FROM EMPLOYEE
WHERE JN_DT > TO_DATE('2010-01-01', 'YYYY-MM-DD');
```

EMP_C ODE	EMP_N AME	DEPT_ CODE	DESIG_ CODE	SE X	ADDRES S	CITY	STATE	PIN	BAS IC	JN_ DT
E001	TAPAN	DP01	D001	M	BALARA MPUR	PURU LIA	WESTBE NGAL	723 103	800 00	21- JAN -24
E002	ROHIT	DP02	D002	M	ABC	PURU LIA	WEST BENGAL	723 102	700 00	21- JAN -24
E004	AMIT	DP03	D003	M	ABC	PURU LIA	WEST BENGAL	723 102	0	21- JAN -24

E005	ABC	DP03	D004	M	ABC_D	PURU LIA	WEST BENGAL	723 102	0	01- JAN -24
E006	DEF	DP02	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	02- JAN -24
E007	GHI	DP01	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	22- JAN -24
E008	ANITA	DP01	D005	F	ABC	PURU LIA	WEST BENGAL	723 102	600 00	23- JAN -24
E009	RITA	DP02	D002	F	ABC	KOLK ATA	WEST BENGAL	723 102	700 00	21- JAN -23
E010	ANKIT A	DP04	D003	F	ABC	KOLK ATA	WEST BENGAL	700 000	950 00	22- MAY -23
E011	MARY	DP03	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	650 00	22- AUG -22
E012	MARY	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	690 00	23- AUG -23

15. Find, how many employees have joined in the month of January?

```
SELECT COUNT(*) AS COUNT
FROM EMPLOYEE
WHERE TO_CHAR(JN_DT, 'MM') = '01';
```


COUNT
9

16. Find the maximum & minimum Basic.

SELECT MAX(BASIC) AS MAXIMUM, MIN(BASIC) AS MINIMUM FROM EMPLOYEE;

MAXIMUM	MINIMUM
95000	0

17. Find how many Female employees are there?

SELECT COUNT(*)
FROM EMPLOYEE
WHERE SEX = 'F';

COUNT (*)
8

18. Replace CITY with existing value converted into uppercase for all rows.

UPDATE EMPLOYEE SET CITY = UPPER(CITY);

19. Find in how many different cities various employees are residing?

SELECT COUNT(DISTINCT CITY) AS DISTINCT_CITY
FROM EMPLOYEE;

DISTINCT_CITY
2

20. Display the employee information in the ascending order of DEPT_CODE and with Department, it should be in the descending order of BASIC.

```
SELECT *
FROM EMPLOYEE
ORDER BY DEPT_CODE ASC, BASIC DESC;
```

EMP_C ODE	EMP_N AME	DEPT_ CODE	DESIG_ CODE	SE X	ADDRES S	CITY	STATE	PIN	BAS IC	JN_ DT
E001	TAPAN	DP01	D001	M	BALARA MPUR	PURU LIA	WESTBE NGAL	723 103	800 00	21- JAN -24
E014	EDIE	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	900 000	690 00	02- JAN -00
E013	EDNA	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	690 00	23- AUG -00
E012	MARY	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	690 00	23- AUG -23
E008	ANITA	DP01	D005	F	ABC	PURU LIA	WEST BENGAL	723 102	600 00	23- JAN -24
E007	GHI	DP01	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	22- JAN -24
E002	ROHIT	DP02	D002	M	ABC	PURU LIA	WEST BENGAL	723 102	700 00	21- JAN -24
E009	RITA	DP02	D002	F	ABC	KOLK ATA	WEST BENGAL	723 102	700 00	21- JAN -23
E015	MAX	DP02	D003	M	ABC	KOLK ATA	WEST BENGAL	900 000	690 00	02- SEP -00

E016	JAMES	DP02	D002	M	ABC	KOLK ATA	WEST BENGAL	900 000	600 00	02- SEP -00
E006	DEF	DP02	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	02- JAN -24
E018	DONA	DP03	D001	F	ABC	KOLK ATA	WEST BENGAL	700 000	890 00	21- SEP -00
E011	MARY	DP03	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	650 00	22- AUG -22
E005	ABC	DP03	D004	M	ABC_D	PURU LIA	WEST BENGAL	723 102	0	01- JAN -24
E004	AMIT	DP03	D003	M	ABC	PURU LIA	WEST BENGAL	723 102	0	21- JAN -24
E010	ANKIT A	DP04	D003	F	ABC	KOLK ATA	WEST BENGAL	700 000	950 00	22- MAY -23
E017	JAMIE	DP04	D002	M	ABC	KOLK ATA	WEST BENGAL	900 000	600 00	24- SEP -00

ASSIGNMENT 2

1. From the EMP table show the minimum, maximum and average basic for each department (show dept. Code).

```
SELECT DEPT_CODE, MAX(BASIC) AS MAXIMUM, MIN(BASIC) AS MINIMUM, AVG(BASIC)
AS AVERAGE FROM EMPLOYEE GROUP BY DEPT_CODE;
```

DEPT_CODE	MAXIMUM	MINIMUM	AVERAGE
-----------	---------	---------	---------

DP01	PURULIA	3
DP02	KOLKATA	3
DP02	PURULIA	2
DP03	KOLKATA	2
DP03	PURULIA	2
DP04	KOLKATA	2

4. Show the designation wise no of employees who have joined in the year 2000 in each department. The listing should appear in the ascending order of no. of employees.

```
SELECT DEPT_CODE, DESIG_CODE, COUNT(*) AS NO_OF_EMPLOYEE FROM
EMPLOYEE WHERE EXTRACT(YEAR FROM JN_DT)=2000 GROUP BY DEPT_CODE,
DESIG_CODE ORDER BY NO_OF_EMPLOYEE;
```

DEPT_CODE	DESIG_CODE	NO_OF_EMPLOYEE
DP04	D002	1
DP02	D003	1
DP02	D002	1

DP03	D001	1
DP01	D005	2

5. Find the department code wise total basic of male employees only for the departments for which such total is more than 50,000 and the listing should appear in the descending order of total basic.

```
SELECT DEPT_CODE, SUM(BASIC) AS TOTAL FROM EMPLOYEE WHERE SEX = 'M'
GROUP BY DEPT_CODE HAVING SUM(BASIC)>50000 ORDER BY TOTAL DESC;
```

DEPT_CODE	TOTAL
DP02	199000
DP01	80000
DP04	60000

6. Show the employee name, Designation description and basic for all employees.

```
SELECT EMPLOYEE.EMP_NAME, DESIGNATION.DESIG_DESE, EMPLOYEE.BASIC FROM
EMPLOYEE JOIN DESIGNATION ON EMPLOYEE.DESIG_CODE =
DESIGNATION.DESIG_CODE;
```

EMP_NAME	DESIG_DESE	BASIC
AMIT	Officer	0
ABC	Clerk	0

DEF	Helper	0
GHI	Helper	0
ANITA	Helper	60000
RITA	Executive	70000
ANKITA	Officer	95000
MARY	Helper	65000
MARY	Helper	69000
EDNA	Helper	69000
EDIE	Helper	69000
MAX	Officer	69000
JAMES	Executive	60000
JAMIE	Executive	60000
DONA	Manager	89000
TAPAN	Manager	80000

ROHIT	Executive	70000
-------	-----------	-------

7. Show the employee name, Designation description, Department Name & Basic for all employees.

```
SELECT EMPLOYEE.EMP_NAME,
       DESIGNATION.DESIG_DESE,
       DEPARTMENT.DEPT_NAME,
       EMPLOYEE.BASIC
FROM EMPLOYEE
JOIN DESIGNATION ON EMPLOYEE.DESIG_CODE = DESIGNATION.DESIG_CODE
JOIN DEPARTMENT ON EMPLOYEE.DEPT_CODE = DEPARTMENT.DEPT_CODE ;
```

EMP_NAME	DESIG_DESE	DEPT_NAME	BASIC
GHI	Helper	Personnel	0
ANITA	Helper	Personnel	60000
MARY	Helper	Personnel	69000
EDNA	Helper	Personnel	69000
EDIE	Helper	Personnel	69000
TAPAN	Manager	Personnel	80000
DEF	Helper	Production	0
RITA	Executive	Production	70000

MAX	Officer	Production	69000
JAMES	Executive	Production	60000
ROHIT	Executive	Production	70000
AMIT	Officer	Purchase	0
ABC	Clerk	Purchase	0
MARY	Helper	Purchase	65000
DONA	Manager	Purchase	89000
ANKITA	Officer	Finnance	95000
JAMIE	Executive	Finnance	60000

8. Find the department Codes in which no employee works.

```
SELECT DEPARTMENT.DEPT_CODE
FROM DEPARTMENT
LEFT JOIN EMPLOYEE ON DEPARTMENT.DEPT_CODE=EMPLOYEE.DEPT_CODE
WHERE EMPLOYEE.DEPT_CODE IS NULL;
```

DEPT_CODE

DP05

9. Find the department names where at least one employee works.

```
SELECT DISTINCT DEPARTMENT.DEPT_NAME, DEPARTMENT.DEPT_CODE
FROM DEPARTMENT
LEFT JOIN EMPLOYEE ON EMPLOYEE.DEPT_CODE = DEPARTMENT.DEPT_CODE
WHERE EMPLOYEE.DEPT_CODE IS NOT NULL;
```

DEPT_NAME	DEPT_CODE
Purchase	DP03
Finnance	DP04
Production	DP02
Personnel	DP01

10. Find the department names where at least 10 employees work.

```
SELECT DEPARTMENT.DEPT_NAME
FROM DEPARTMENT
INNER JOIN EMPLOYEE ON DEPARTMENT.DEPT_CODE = EMPLOYEE.DEPT_CODE
GROUP BY DEPT_NAME HAVING COUNT(EMPLOYEE.DEPT_CODE) >= 10;
```

DEPT_NAME
Personnel

11. Find the department code in which employee with highest Basic works.

```
SELECT EMPLOYEE.DEPT_CODE , EMPLOYEE.BASIC
FROM EMPLOYEE
WHERE EMPLOYEE.BASIC = (SELECT MAX(BASIC) FROM EMPLOYEE);
```

DEPT_CODE	BASIC
DP04	95000

12. Find the Designation description of the employee with highest basic.

```
SELECT DESIGNATION.DESIG_DESE
FROM EMPLOYEE
JOIN DESIGNATION ON EMPLOYEE.DESIG_CODE = DESIGNATION.DESIG_CODE
WHERE EMPLOYEE.BASIC = (SELECT MAX(BASIC) FROM EMPLOYEE);
```

DESIG_DESE
Officer

13. Find the no. of managers in each department.

```
SELECT DEPARTMENT.DEPT_NAME, COUNT(EMPLOYEE.EMP_CODE) AS MANAGERS
FROM DEPARTMENT
INNER JOIN EMPLOYEE ON DEPARTMENT.DEPT_CODE= EMPLOYEE.DEPT_CODE
WHERE EMPLOYEE.DESIG_CODE = 'D001'
GROUP BY DEPARTMENT.DEPT_NAME;
```

DEPT_NAME	MANAGERS
Personnel	1
Purchase	1

14. Find the maximum basic from EMP table without using MAX().
SELECT BASIC AS MAX_BASIC FROM EMPLOYEE
ORDER BY BASIC DESC
FETCH FIRST 1 ROWS ONLY;

MAX_BASIC
95000

15. Find the minimum basic from EMP table without using MIN().
SELECT BASIC AS MIN_BASIC FROM EMPLOYEE
ORDER BY BASIC ASC
FETCH FIRST 1 ROWS ONLY;

MIN_BASIC
0

16. Find the name of the department with the highest total basic. Do the same for the highest average basic and maximum no. of employee.

SELECT DEPT_NAME, SUM(BASIC) AS Total_Basic
FROM EMPLOYEE E
JOIN DEPARTMENT D ON E.DEPT_CODE = D.DEPT_CODE
GROUP BY DEPT_NAME
ORDER BY Total_Basic DESC
FETCH FIRST 1 ROWS ONLY;

DEPT_NAME	TOTAL_BASIC
Personnel	347000

SELECT DEPT_NAME, AVG(BASIC) AS Average_Basic
FROM EMPLOYEE E
JOIN DEPARTMENT D ON E.DEPT_CODE = D.DEPT_CODE
GROUP BY DEPT_NAME
ORDER BY Average_Basic DESC
FETCH FIRST 1 ROWS ONLY;

DEPT_NAME	AVERAGE_BASIC
Finnance	77500

```

SELECT DEPT_NAME, COUNT(EMP_CODE) AS Num_Employees
FROM EMPLOYEE E
JOIN DEPARTMENT D ON E.DEPT_CODE = D.DEPT_CODE
GROUP BY DEPT_NAME
ORDER BY Num_Employees DESC
FETCH FIRST 1 ROWS ONLY;

```

DEPT_NAME	NUM_EMPLOYEES
Personnel	6

17. Insert same rows into EMP table with designation code not existing in DESIGNATION table

```

insert into EMPLOYEE values ('E0019','ANI','DP01', 'D009', 'F', 'ABC','PURULIA','WEST
BENGAL','723102',68000, to_date('23:01:2002','dd:mm:yyyy'));

```

```

insert into EMPLOYEE values ('E0020','ANKIT','DP02', 'D017', 'M', 'ABC','PURULIA','WEST
BENGAL','723102',68000, to_date('23:01:2000','dd:mm:yyyy'));

```

```

insert into EMPLOYEE values ('E0021','DEEP','DP02', 'D016', 'M', 'ABC','PURULIA','WEST
BENGAL','723102',68000, to_date('23:01:2000','dd:mm:yyyy'));

```

```

insert into EMPLOYEE values ('E0022','DEEPA','DP01', 'D016', 'F', 'ABC','PURULIA','WEST
BENGAL','723102',68000, to_date('23:01:2000','dd:mm:yyyy'));

```

EMP_C ODE	EMP_N AME	DEPT_ CODE	DESIG_ CODE	SE X	ADDRES S	CITY	STATE	PIN	BAS IC	JN_ DT
E0019	ANI	DP01	D009	F	ABC	PURU LIA	WEST BENGAL	723 102	680 00	23- JAN -02
E0020	ANKIT	DP02	D017	M	ABC	PURU LIA	WEST BENGAL	723 102	680 00	23- JAN -00
E0021	DEEP	DP02	D016	M	ABC	PURU LIA	WEST BENGAL	723 102	680 00	23- JAN -00

E0022	DEEPA	DP01	D016	F	ABC	PURU LIA	WEST BENGAL	723 102	680 00	23- JAN -00
E004	AMIT	DP03	D003	M	ABC	PURU LIA	WEST BENGAL	723 102	0	21- JAN -24
E005	ABC	DP03	D004	M	ABC_D	PURU LIA	WEST BENGAL	723 102	0	01- JAN -24
E006	DEF	DP02	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	02- JAN -24
E007	GHI	DP01	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	22- JAN -24
E008	ANITA	DP01	D005	F	ABC	PURU LIA	WEST BENGAL	723 102	600 00	23- JAN -24
E009	RITA	DP02	D002	F	ABC	KOLK ATA	WEST BENGAL	723 102	700 00	21- JAN -23
E010	ANKIT A	DP04	D003	F	ABC	KOLK ATA	WEST BENGAL	700 000	950 00	22- MAY -23
E011	MARY	DP03	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	650 00	22- AUG -22
E012	MARY	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	690 00	23- AUG -23
E013	EDNA	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	690 00	23- AUG -00
E014	EDIE	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	900 000	690 00	02- JAN -00

E015	MAX	DP02	D003	M	ABC	KOLK ATA	WEST BENGAL	900 000	690 00	02- SEP -00
E016	JAMES	DP02	D002	M	ABC	KOLK ATA	WEST BENGAL	900 000	600 00	02- SEP -00
E017	JAMIE	DP04	D002	M	ABC	KOLK ATA	WEST BENGAL	900 000	600 00	24- SEP -00
E018	DONA	DP03	D001	F	ABC	KOLK ATA	WEST BENGAL	700 000	890 00	21- SEP -00
E001	TAPAN	DP01	D001	M	BALARA MPUR	PURU LIA	WESTBE NGAL	723 103	800 00	21- JAN -24
E002	ROHIT	DP02	D002	M	ABC	PURU LIA	WEST BENGAL	723 102	700 00	21- JAN -24

18. Delete the rows from EMP table with invalid DESIG_CODE.

```
DELETE FROM EMPLOYEE E
WHERE NOT EXISTS(
SELECT *
FROM DESIGNATION
WHERE DESIG_CODE=E.DESIG_CODE
);
```

EMP_C ODE	EMP_N AME	DEPT_ CODE	DESIG_ CODE	SE X	ADDRES S	CITY	STATE	PIN	BAS IC	JN_ DT
E004	AMIT	DP03	D003	M	ABC	PURU LIA	WEST BENGAL	723 102	0	21- JAN -24
E005	ABC	DP03	D004	M	ABC_D	PURU LIA	WEST BENGAL	723 102	0	01- JAN -24
E006	DEF	DP02	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	02- JAN -24

E007	GHI	DP01	D005	M	ABC	PURU LIA	WEST BENGAL	723 102	0	22- JAN -24
E008	ANITA	DP01	D005	F	ABC	PURU LIA	WEST BENGAL	723 102	600 00	23- JAN -24
E009	RITA	DP02	D002	F	ABC	KOLK ATA	WEST BENGAL	723 102	700 00	21- JAN -23
E010	ANKIT A	DP04	D003	F	ABC	KOLK ATA	WEST BENGAL	700 000	950 00	22- MAY -23
E011	MARY	DP03	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	650 00	22- AUG -22
E012	MARY	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	690 00	23- AUG -23
E013	EDNA	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	700 000	690 00	23- AUG -00
E014	EDIE	DP01	D005	F	ABC	KOLK ATA	WEST BENGAL	900 000	690 00	02- JAN -00
E015	MAX	DP02	D003	M	ABC	KOLK ATA	WEST BENGAL	900 000	690 00	02- SEP -00
E016	JAMES	DP02	D002	M	ABC	KOLK ATA	WEST BENGAL	900 000	600 00	02- SEP -00
E017	JAMIE	DP04	D002	M	ABC	KOLK ATA	WEST BENGAL	900 000	600 00	24- SEP -00
E018	DONA	DP03	D001	F	ABC	KOLK ATA	WEST BENGAL	700 000	890 00	21- SEP -00

E001	TAPAN	DP01	D001	M	BALARA MPUR	PURU LIA	WEST BENGAL	723 103	800 00	21- JAN -24
E002	ROHIT	DP02	D002	M	ABC	PURU LIA	WEST BENGAL	723 102	700 00	21- JAN -24

19. Find the name of the female employees with basic greater than the average basic of their respective department.

```
SELECT EMP_NAME, BASIC, DEPT_CODE
FROM EMPLOYEE E1
WHERE SEX='F' AND
BASIC>(
SELECT AVG(BASIC)
FROM EMPLOYEE E2
WHERE E1.DEPT_CODE = E2.DEPT_CODE
GROUP BY E2.DEPT_CODE
)ORDER BY DEPT_CODE ;
```

EMP_NAME	BASIC	DEPT_CODE
EDNA	69000	DP01
EDIE	69000	DP01
ANITA	60000	DP01
MARY	69000	DP01
RITA	70000	DP02
DONA	89000	DP03
MARY	65000	DP03
ANKITA	95000	DP04

20. Find the number of female managers.

```
SELECT COUNT(E.EMP_NAME) AS NO_OF_FEMALE_MANAGER
FROM EMPLOYEE E JOIN DESIGNATION D
ON E.DESIG_CODE=D.DESIG_CODE WHERE E.SEX='F' AND D.DESIG_DESE='Manager';
```

NO_OF_FEMALE_MANAGER

1

ASSIGNMENT 3

Q1. In an organization, number of departments exists. Each department has a name & unique code. Number of employees work in each department. Each employee has unique employee code. Detailed information like name, address, city, basic, date of join are also stored. In a leave register for each employee leave records are kept showing leave type (CL/EL/ML etc.), from-date and to-date. When an employee retires or resigns then all the leave information pertaining to him are also deleted. Basic salary must be within Rs.5000 to Rs.9000. A department can not be deleted if any employee record refers to it. Valid grades are A/B/C. Employee name must be in uppercase only. Default value for joining date is system date. Design & implement the tables with necessary constraints to support the scenario depicted above.

SQL:

```
CREATE TABLE DEPARTMENT_3(  
    DEPT_CODE CHAR(5) PRIMARY KEY,  
    DEPT_NAME CHAR(15)  
);
```

```
CREATE TABLE EMP_3(  
    EMP_CODE CHAR(5) PRIMARY KEY,  
    EMP_NAME CHAR(50) CHECK (EMP_NAME = UPPER(EMP_NAME)),  
    GRADE CHAR(1) CHECK (GRADE IN ('A', 'B', 'C')),  
    DEPT_CODE CHAR(5) REFERENCES DEPARTMENT_3,  
    ADDRESS CHAR(100),  
    CITY CHAR(50),  
    BASIC NUMBER CHECK(BASIC BETWEEN 5000 AND 9000),  
    DJN DATE DEFAULT SYSDATE NOT NULL  
);
```

```
CREATE TABLE LEAVE_3(  
    EMP_CODE CHAR(5),  
    L_TYPE CHAR(2) CHECK (L_TYPE IN ('CL','EL','ML')),  
    F_DT DATE,  
    T_DT DATE,  
    FOREIGN KEY(EMP_CODE) REFERENCES EMP_3(EMP_CODE) ON DELETE  
CASCADE  
);
```

```
insert into DEPARTMENT_3 VALUES('DP01','FINANCE');  
insert into DEPARTMENT_3 VALUES('DP02','HR');  
insert into DEPARTMENT_3 VALUES('DP03','SALES');  
insert into DEPARTMENT_3 VALUES('DP04','PRODUCTION');
```

SELECT * FROM DEPARTMENT_3;

DEPT_CODE	DEPT_NAME
DP01	FINANCE
DP02	HR
DP03	SALES
DP04	PRODUCTION

insert into EMP_3 values ('E001','ANIKET','A', 'DP01','PURULIA','WEST BENGAL',5000,
to_date('23:01:2024','dd:mm:yyyy'));

insert into EMP_3(EMP_CODE, EMP_NAME, GRADE, DEPT_CODE, ADDRESS, CITY,
BASIC) values ('E002','DIP','A', 'DP02','KOLKATA','WEST BENGAL',5800);

insert into EMP_3 values ('E003','ANKITA','B', 'DP02','KOLKATA','WEST BENGAL',6000,
to_date('23:01:2000','dd:mm:yyyy'));

insert into EMP_3 values ('E005','JOHN','C', 'DP03','KOLKATA','WEST BENGAL',8000,
to_date('23:01:2000','dd:mm:yyyy'));

insert into EMP_3 values ('E006','JOHNY','C', 'DP04','KOLKATA','WEST BENGAL',9000,
to_date('23:01:2000','dd:mm:yyyy'));

insert into EMP_3 values ('E007','JANE','B', 'DP03','KOLKATA','WEST BENGAL',9000,
to_date('23:04:2004','dd:mm:yyyy'));

insert into EMP_3 values ('E008','JAKE','C', 'DP01','KOLKATA','WEST BENGAL',9000,
to_date('23:04:2004','dd:mm:yyyy'));

insert into EMP_3 values ('E009','JAY','A', 'DP02','KOLKATA','WEST BENGAL',9000,
to_date('02:04:2003','dd:mm:yyyy'));

insert into EMP_3 values ('E010','FIONA','A', 'DP03','KOLKATA','WEST BENGAL',9000,
to_date('02:04:2003','dd:mm:yyyy'));

insert into EMP_3 values ('E011','FIONA','B', 'DP04','KOLKATA','WEST BENGAL',9000,
to_date('02:04:2003','dd:mm:yyyy'));

SELECT * from EMP_3;

EMP_CODE	EMP_NAME	GRADE	DEPT_CODE	ADDRESS	CITY	BASIC	DJN
E003	ANKITA	B	DP02	KOLKATA	WEST BENGAL	6000	23-JAN-00
E005	JOHN	C	DP03	KOLKATA	WEST BENGAL	8000	23-JAN-00
E006	JOHNY	C	DP04	KOLKATA	WEST BENGAL	9000	23-JAN-00
E007	JANE	B	DP03	KOLKATA	WEST BENGAL	9000	23-APR-04
E008	JAKE	C	DP01	KOLKATA	WEST BENGAL	9000	23-APR-04

E009	JAY	A	DP02	KOLKATA	WEST BENGAL	9000	02-APR-03
E010	FIONA	A	DP03	KOLKATA	WEST BENGAL	9000	02-APR-03
E011	FIONA	B	DP04	KOLKATA	WEST BENGAL	9000	02-APR-03
E001	ANIKET	A	DP01	PURULIA	WEST BENGAL	5000	23-JAN-24
E002	DIP	A	DP02	KOLKATA	WEST BENGAL	5800	28-MAY-24

```
insert into LEAVE_3 values('E001', 'CL',to_date('23:08:2023', 'dd:mm:yyyy'),
to_date('25:08:2023', 'dd:mm:yyyy'));
insert into LEAVE_3 values('E001', 'ML',to_date('23:08:2023', 'dd:mm:yyyy'),
to_date('25:02:2024', 'dd:mm:yyyy'));
```

SELECT * from LEAVE_3;

EMP_CODE	L_TYPE	F_DT	T_DT
E001	CL	23-AUG-23	25-AUG-23
E001	ML	23-AUG-23	25-FEB-24

Q2. Try to violate the constraints that you have implemented in the table & note, what happens. [Try with suitable INSERT/UPDATE/DELETE instruction].

i. Insert with a Lowercase EMP_NAME

```
INSERT INTO EMP_3 VALUES ('E012', 'john', 'C', 'DP03', 'KOLKATA', 'WEST BENGAL', 8000,
TO_DATE('23:01:2000', 'DD:MM:YYYY'));
```

ORA-02290: check constraint

(SQL_BEEGUPFRNJMVCMRTYDAPFCUP.SYS_C00159825076) violated ORA-06512: at "SYS.DBMS_SQL", line 1721

ii. Insert with Invalid EMP_CODE in LEAVE_3

```
INSERT INTO LEAVE_3 VALUES ('E999', 'CL', TO_DATE('23:08:2023', 'DD:MM:YYYY'),
TO_DATE('25:08:2023', 'DD:MM:YYYY'));
```

ORA-02290: check constraint

(SQL_BEEGUPFRNJMVCMRTYDAPFCUP.SYS_C00159825076) violated ORA-06512: at "SYS.DBMS_SQL", line 1721

Q3.

a) create a view showing employee code, name, dcode & Basic For a particular department.

b)Try to ensure a row into the view with valid department & also with invalid ones.

c)Find the newly inserted row in the table From which view was created .

d) Try to increment basic by Rs.100/-

e) Check it in the original table.

f) Delete the view.

SQL:

create view VIEW3

as select e.emp_code, e.emp_name, e.basic, d.dept_code

from EMP_3 e, DEPARTMENT_3 d

where e.dept_code=d.dept_code;

EMP_CODE	EMP_NAME	BASIC	DEPT_CODE
E001	ANIKET	5000	DP01
E002	DIP	5800	DP02
E003	ANKITA	6000	DP02
E005	JOHN	8000	DP03
E006	JOHNY	9000	DP04
E007	JANE	9000	DP03
E008	JAKE	9000	DP01
E009	JAY	9000	DP02
E010	FIONA	9000	DP03
E011	FIONA	9000	DP04

insert into VIEW3 values('E012', 'ARKA', 7000, 'DP01');

EMP_CODE	EMP_NAME	BASIC	DEPT_CODE
E001	ANIKET	5000	DP01
E002	DIP	5800	DP02
E003	ANKITA	6000	DP02
E005	JOHN	8000	DP03
E006	JOHNY	9000	DP04
E007	JANE	9000	DP03
E008	JAKE	9000	DP01
E009	JAY	9000	DP02
E010	FIONA	9000	DP03
E011	FIONA	9000	DP04
E012	ARKA	7000	DP01

UPDATE view3

SET BASIC = BASIC + 100

WHERE BASIC < 9000;

EMP_CODE	EMP_NAME	BASIC	DEPT_CODE
E001	ANIKET	5200	DP01
E002	DIP	6000	DP02

E003	ANKITA	6200	DP02
E005	JOHN	8200	DP03
E006	JOHNY	9000	DP04
E007	JANE	9000	DP03
E008	JAKE	9000	DP01
E009	JAY	9000	DP02
E010	FIONA	9000	DP03
E011	FIONA	9000	DP04
E012	ARKA	7200	DP01

SELECT * FROM EMP_3;

EMP_CO DE	EMP_NAME	BASIC	DEPT_CO DE
E001	ANIKET	5200	DP01
E002	DIP	6000	DP02
E003	ANKITA	6200	DP02
E005	JOHN	8200	DP03
E006	JOHNY	9000	DP04
E007	JANE	9000	DP03
E008	JAKE	9000	DP01

E009	JAY	9000	DP02
E010	FIONA	9000	DP03
E011	FIONA	9000	DP04
E012	ARKA	7200	DP01

DROP view VIEW3;

Q4.

a) create a view Showing empcode, name, deptname, basic, leave type, From date & to date.

b) Try to insert a row in the view. Check what happens?

c) Try to increment basic by Rs.100.

d) Delete the view.

create view VIEW4

as select e.emp_code, e.emp_name, e.basic, d.dept_name, l.l_type, l.f_dt, l.t_dt
from leave_3 l, emp_3 e, department_3 d
where l.emp_code=e.emp_code and e.dept_code=d.dept_code;

EMP_CODE	EMP_NAME	BASIC	DEPT_NAME	L_TYPE	F_DT	T_DT
E001	ANIKET	5200	FINANCE	CL	23-AUG-23	25-AUG-23
E001	ANIKET	5200	FINANCE	ML	23-AUG-23	25-FEB-24

INSERT INTO VIEW4 VALUES('E005','JOHN',8100,'SALES','EL',to_date('22:02:2024','dd:mm:yyyy'), to_date('27:02:2024','dd:mm:yyyy'));

ORA-01779: cannot modify a column which maps to a non key-preserved table

UPDATE view4

SET BASIC = BASIC + 100;

ORA-01779: cannot modify a column which maps to a non key-preserved table

DROP view VIEW4;

Q5) a) Create a table having empcode , Name, deptname, & basic From the existing tables along with the

records of the employee who are in a particular department (say, d1) and with a basic Rs. 7000/-

b) From the existing table, add the employees with the basic salary greater than or equal to 7000/-

c) Alter the table to add a net pay column.

d) Replace net pay with 1.5* Basic.

e) Try to remove the net pay column.

```
CREATE TABLE NEWEMPTABLE AS
(SELECT E.EMP_CODE, E.EMP_NAME, E.BASIC, D.DEPT_CODE, D.DEPT_NAME
  FROM EMP_3 E, DEPARTMENT_3 D
  WHERE E.DEPT_CODE=D.DEPT_CODE AND E.BASIC>=7000 AND
D.DEPT_CODE='DP01');
SELECT * from newEMPtable;
```

EMP_CODE	EMP_NAME	BASIC	DEPT_CODE	DEPT_NAME
E008	JAKE	9000	DP01	FINANCE
E012	ARKA	7200	DP01	FINANCE

```
alter table newEMPtable
add NET_PAY number(10, 2);
```

```
update newEMPtable
set NET_PAY= 1.5 * BASIC
where BASIC>0;
```

```
select * from newEMPtable;
```

EMP_CODE	EMP_NAME	BASIC	DEPT_CODE	DEPT_NAME	NET_PAY
E008	JAKE	9000	DP01	FINANCE	13500
E012	ARKA	7200	DP01	FINANCE	10800

```
ALTER TABLE newEMPtable
DROP COLUMN NET_PAY;
```

Q6.

Drop all the tables that you have created.

SQL:

Drop table DEPARTMENT_3;

Drop table EMP_3;

Drop table newEMPtable;

Drop table LEAVE_3;

ASSIGNMENT 4

- 1 a) Create EMP table with ECODE(primary key), ENAME, DCODE, GRADE, BASIC & JN-DT as the columns.[except BASIC & JN-DT,all columns are of char type and size of Grade is 1.]
b)Insert number of rows.

```
CREATE TABLE EMP_4(  
    ECODE CHAR(5) PRIMARY KEY,  
    ENAME CHAR(50),  
    GRADE CHAR(1) ,  
    DCODE CHAR(5),  
    BASIC NUMBER ,  
    JN_DT DATE  
);
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES  
('001', 'John', 'A', 'D001', 50000, TO_DATE('2024-01-01', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES ('002', 'Alice', 'B', 'D002', 45000, TO_DATE('2023-05-15', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES ('003', 'Bob', 'C', 'D001', 40000, TO_DATE('2024-02-20', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES ('004', 'Emma', 'A', 'D003', 55000, TO_DATE('2022-11-10', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES ('005', 'Michael', 'B', 'D002', 47000, TO_DATE('2024-04-10', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES ('006', 'Sophia', 'A', 'D001', 52000, TO_DATE('2023-09-25', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES ('007', 'William', 'B', 'D002', 48000, TO_DATE('2024-03-12', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES ('008', 'Olivia', 'C', 'D003', 42000, TO_DATE('2023-11-05', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES ('009', 'James', 'A', 'D001', 53000, TO_DATE('2022-07-20', 'YYYY-MM-DD'));
```

```
INSERT INTO EMP_4 (ECODE, ENAME, GRADE, DCODE, BASIC, JN_DT)  
VALUES ('010', 'Emma', 'B', 'D002', 46000, TO_DATE('2024-01-08', 'YYYY-MM-DD'));
```

2. change the column heading as shown below, So that in subsequent SELECT statement newly set heading will be shown:

ECODE	EMPLOYEE CODE
ENAME	NAME
DCODE	DEPT.CODE
JN-DT	JOINING DATE

```
ALTER TABLE EMP_4
RENAME COLUMN ECODE TO "EMPLOYEE CODE";
```

```
ALTER TABLE EMP_4
RENAME COLUMN ENAME TO NAME;
```

```
ALTER TABLE EMP_4
RENAME COLUMN JN_DT TO "JOINING DATE";
```

```
ALTER TABLE EMP_4
RENAME COLUMN DCODE TO "DEPTCODE";
```

2 (can be done this way too!)

```
COLUMN ECODE FORMAT A20 HEADING "EMPLOYEE CODE";
```

```
COLUMN ENAME FORMAT A20 HEADING NAME;
```

```
COLUMN JN_DT FORMAT A20 HEADING "JOINING DATE";
```

```
COLUMN DCODE FORMAT A20 HEADING DEPTCODE;
```

```
COLUMN GRADE FORMAT A20 HEADING GRADE;
```

-- selecting some tuples

```
select *
from EMP_4;
```

-- =====

EMPLOYEE CODE	NAME	GRADE	DEPTCODE	BASIC	JOINING DATE
001	John	A	D001	50,000	01-JAN-24
002	Alice	B	D002	45,000	15-MAY-23
003	Bob	C	D001	40,000	20-FEB-24
004	Emma	A	D003	55,000	10-NOV-22
005	Michael	B	D002	47,000	10-APR-24
006	Sophia	A	D001	52,000	25-SEP-23
007	William	B	D002	48,000	12-MAR-24
008	Olivia	C	D003	42,000	05-NOV-23
009	James	A	D001	53,000	20-JUL-22
010	Emma	B	D002	46,000	08-JAN-24

10 rows selected.

3. Set the format of columns as mentioned below, So that in subsequent SELECT statement ,values appear in the specified format:

- *format of BASIC is such that a value of 7000 will be shown as 7,000
- *Format of GRADE will be such that full column name appears in the display.
- *For JN-DT format is such that 01-JAN-00 will be shown as JANURY

01,2000

COLUMN BASIC FORMAT 99,990;

EMPLOYEE CODE	NAME	GRADE	DEPTCODE	BASIC	JOINING DATE
001	John	A	D001	50,000	01-JAN-24
002	Alice	B	D002	45,000	15-MAY-23
003	Bob	C	D001	40,000	20-FEB-24

```
SELECT ECODE, ENAME, CONCAT('GRADE:', GRADE), BASIC,
TO_CHAR("JN_DT", 'fmMonth DD,YYYY') FROM EMP_4;
```

```
SQL> SELECT ECODE, ENAME, CONCAT('GRADE:', GRADE), BASIC, TO_CHAR("JN_DT", 'fmMonth DD,YYYY') FROM EMP_4;
```

EMPLOYEE CODE	NAME	CONCAT(	BASIC	TO_CHAR("JN_DT", 'FMMONTHDD,YYYY')
001	John	GRADE:A	50,000	January 1,2024
002	Alice	GRADE:B	45,000	May 15,2023
003	Bob	GRADE:C	40,000	February 20,2024
004	Emma	GRADE:A	55,000	November 10,2022
005	Michael	GRADE:B	47,000	April 10,2024
006	Sophia	GRADE:A	52,000	September 25,2023
007	William	GRADE:B	48,000	March 12,2024
008	Olivia	GRADE:C	42,000	November 5,2023
009	James	GRADE:A	53,000	July 20,2022
010	Emma	GRADE:B	46,000	January 8,2024

10 rows selected.

- 4 a) Show the display attributes of all the columns.
- b) Show the display attributes of particular column.
- c) Suppress the newly set attributes of JN-DT .Try a select statement.
- d) Reset the newly set attributes of JN-DT
- e) Reset the newly set attributes of all columns.
- d) Shown the display attributes of all columns.

-- 4a
COLUMN;

```

SQL> COLUMN;
COLUMN      BASIC ON
FORMAT      99,990

COLUMN      GRADE ON
HEADING      'GRADE'
FORMAT      A20

COLUMN      DCODE ON
HEADING      'DEPTCODE'
FORMAT      A20

COLUMN      JN_DT ON
HEADING      'JOINING DATE'
FORMAT      A20

COLUMN      ENAME ON
HEADING      'NAME'
FORMAT      A20

COLUMN      ECODE ON
HEADING      'EMPLOYEE CODE'
FORMAT      A20

```

-- 4b
COLUMN JN_DT;

```

SQL> COLUMN JN_DT;
COLUMN      JN_DT ON
HEADING      'JOINING DATE'
FORMAT      A20
SQL>

```

-- 4c

```

SELECT ECODE,
       ENAME,
       DCODE,
       JN_DT CLEAR as "JN_DT"
FROM EMP_4;

```

EMPLOYEE CODE	NAME	DEPTCODE	CLEAR
001	John	D001	01-JAN-24
002	Alice	D002	15-MAY-23
003	Bob	D001	20-FEB-24
004	Emma	D003	10-NOV-22
005	Michael	D002	10-APR-24
006	Sophia	D001	25-SEP-23
007	William	D002	12-MAR-24
008	Olivia	D003	05-NOV-23
009	James	D001	20-JUL-22
010	Emma	D002	08-JAN-24

-- 4d

COLUMN JN_DT CLEAR;

-- 4e

CLEAR COLUMNS;

```
SQL> CLEAR COLUMNS;  
columns cleared
```

-- 4f

COLUMN;

```
SQL> COLUMN;  
SP2-0045: * no COLUMN defined  
SQL> |
```

--5 5. a) Show the records from EMP table in the ascending order of DCODE . DCODE value will be shown

-- 5a

```
SELECT ECODE AS "EMPLOYEE CODE",  
       ENAME AS "NAME",  
       CASE WHEN RN = 1 THEN DCODE ELSE NULL END AS "DEPT.CODE",  
       JN_DT AS "JOINING DATE"  
FROM (  
    SELECT ECODE,  
           ENAME,  
           DCODE,  
           JN_DT,  
           ROW_NUMBER() OVER (PARTITION BY DCODE ORDER BY DCODE) AS RN  
    FROM EMP_4  
    ) temp  
ORDER BY DCODE, RN;
```

EMPLOYEE CODE	NAME	DEPT.CODE	JOINING DATE
003	Bob	D001	20-FEB-24
006	Sophia	-	25-SEP-23
001	John	-	01-JAN-24
009	James	-	20-JUL-22
007	William	D002	12-MAR-24
010	Emma	-	08-JAN-24
002	Alice	-	15-MAY-23
005	Michael	-	10-APR-24
004	Emma	D003	10-NOV-22
008	Olivia	-	05-NOV-23